Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

Prestressed Steel Strand (PC Strand)
2-Wire, 3-Wire and 7-Wire Strand

from

Socitrel – Sociedade Industrial de Trefilaria, S.A.

Programme:	The International EPD® System, www.environdec.com
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An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

General information

Programme information

Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
E-mail:	info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): PCR Construction Products (2019:14) v.1.3.4 prEN 17662:2021 Execution of steel structures and aluminium structures - Environmental Product Declarations - Product category rules complementary to EN 15804 for Steel, Iron and Aluminium structural products for use in construction works
PCR review was conducted by: The Technical Committee of the International EPD® System. See www.environdec.com/TC for a list of members. Review chair: Claudia A. Peña, University of Concepción, Chile. The review panel may be contacted via the Secretariat www.environdec.com/contact.
Life Cycle Assessment (LCA)
LCA accountability: Marisa Almeida, Centro Tecnológico da Cerâmica e do Vidro
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ EPD verification by individual verifier Third-party verifier: Rubén Carnerero Acosta, IK Ingeniería S.L. Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☒ Yes ☐ No [Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier]

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

Company information

Owner of the EPD:
SOCITREL – Sociedade Industrial de Trefilaria, S.A.

Contact:
Ana Polidoro (Sustainability) - ana.polidoro@socitrel.pt

Description of the organisation:
A SOCITREL – Sociedade Industrial de Trefilaria, S.A. is a company specialized in the manufacture and trading of steel wire for application in the most diverse fields of activity: industry, agriculture and construction.

Founded in 1971, SOCITREL began with the production of mild steel wire and later evolved to more sophisticated products with a high carbon content, amongst which prestressed steel wire. Now, is a benchmark company in its activity sector, and the only company that produces zinc-plated wire in Portugal, being one of the most modern in Europe.

After several vicissitudes, which naturally fill the pages of more than 50 years of its history, the company is acquired by the Ramada Group in 2015, to which it currently belongs.
Manufacturing high quality products, using the most appropriate cutting-edge technologies, has always been a concern of the company, since its inception. This strategy was reinforced in 2015, through the development of increasingly personalized products.

The proximity established in the relationship with the customer is another factor that distinguishes Socitrel's way of being.

SOCITREL aims to continually modernise itself in the search for new products and new markets, using/developing new technologies for a better performance, so that it can reach a level of excellence in terms of operational, financial and environmental performance and, thus, contribute to the improvement of society.

Product-related or management system-related certifications:
This site that produced the product covered in this EPD have major certifications, including ISO 9001, ISO 14001, ISO 45001, and product conformity.

Name and location of production site(s):
Rua Gago Coutinho, nº 545
4745 – 521 S. Romão do Coronado
Coronado (São Romão e São Mamede), Porto - Portugal

E-mail: geral@socitrel.pt
https://www.socitrel.pt/

Product information

Product name:

Prestressed steel strand (PC strand)

Product identification:

Prestressed steel strand (2, 3 and 7 number of wires) produced by Socitrel.

Product description:

The prestressed steel strand produced by the company SOCITREL is a high tensile strength steel strand used in the manufacture of prestressed and post-tensioned concrete, post-tensioned elements in large supporting structures, in construction and infrastructures. This product can range in 2, 3 and 7 wire steel strands for prestressing and posttensioning of concrete, produced with micro-alloyed steel wire rods that are drawn to smaller dimensions and twisted into strands.

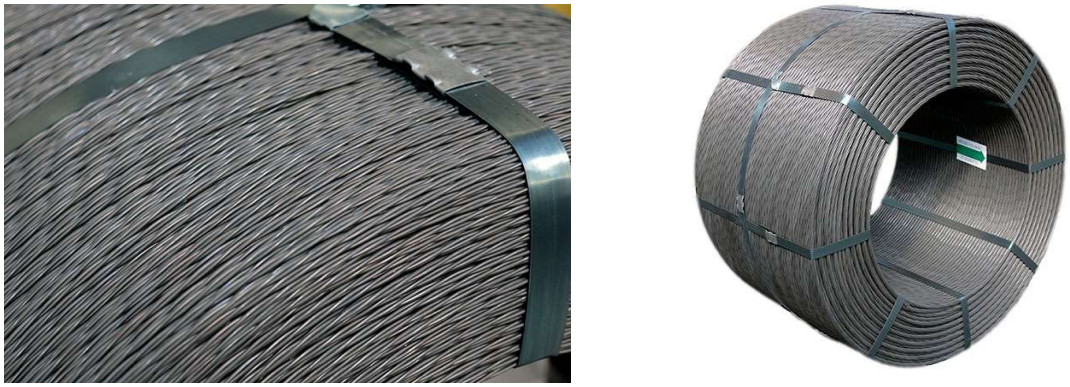


Figure 1 – Examples of prestressed steel strand produced by SOCITREL.

Figure 2 – Examples of prestressed steel strand product usage stage produced by SOCITREL.

This EPD refers to the 2, 3 and 7 wires of prestressed steel strand products, whose characteristics are shown in (Table 1) correspond to the average of the characteristics of prestressed steel strand products according to the number of wires.

Table 1 – Properties of prestressed steel strand products.

Nr. of wires	Diameter (mm)	Tensile Strength (MPa)	Surface
2	4,50	1860	Smooth/ Indented
3	5,20	2060/2160	
7	6,85 9,30 11,30 12,50 12,70 12,90/13,00 15,20 15,30 15,70/16,00	2060/2160 1860 1860 1860 1860 1860 1860 1860 1860	

Application/intended use

The prestressed steel strand products are used for the following applications:

- Alveolar beams;
- Large prestressed elements;
- Structural support.

Specific information on dimension tolerances, constructional data, and mechanical and chemical properties can also be found in the relevant literature and/or the following product standards:

- LNEC E453 Cordões de aço para pré-esforço;
- UNE 36094 Alambres y cordones de acero para armaduras de hormigón pretensado;
- prEN 10138 Prestressing Steels Part 1: General requirements + Part 3 Strand;
- BS 5896 Specification for High Tensile Steel Wire and Strand for the Prestressing of Concrete;
- NF A 35-045 Aciers de Précontrainte – Partie 1 : Prescriptions générales + Partie 3 – Torons;
- SS 212551 Prestressing steels – General requirements + SS 212553 Prestressing steels – Strand;
- ASTM A 416 Standard specification for Steel Strand, Uncoated Seven-Wire for Prestressed Concrete;
- ASTM A 886 Standard specification for Steel Strand, Indented, Seven-Wire Stress-Relieved for Prestressed Concrete;
- NBR 7483: 2008 Steel Strand for Prestressed Concrete Structures, Specification;
- Others according to customer needs.

<u>UN CPC code</u>:
412 Products of iron or steel
421 Structural metal products and parts thereof

<u>Geographical scope:</u>
The dataset used to model the production process is, whenever possible, based on specific Portuguese data, since it is produced in this country, or whenever it is not possible, average European data.

LCA information

Declared unit:

1 ton of prestressed steel strand products

Reference service life:

Not applicable

Time representativeness:

The temporal representativeness is based on the data sets used in the Ecoinvent v3.9.1 database (most recent at the time of the preparation of the EPD), with the update of the Portuguese electric mix to the year 2023. The specific data of the manufacturer refer to specific data production from 2023.

The data set used to model the upstream production processes and the process itself reflect the physical reality and the technology used. For each process/material used in the process modeling, data sets representative of the technologies used were used.

Electric mix:

Portugal mix in 2023: 70.7% renewable energy (29.3% wind power, 26.7% water, 8.1% solar, 6.6% biomass, 2.6% renewable cogeneration and 3.9% other renewable), 22.7% non-renewable energy (22.3% natural gas, 18% combined cycle, 4.4% cogeneration, and 0.4% other thermal, 0.1% cogeneration, 0.4% others), and 6.5% pumped energy (source: REN).
The climate impact of electricity production: 0.0615 kg CO_2 eq./kWh.

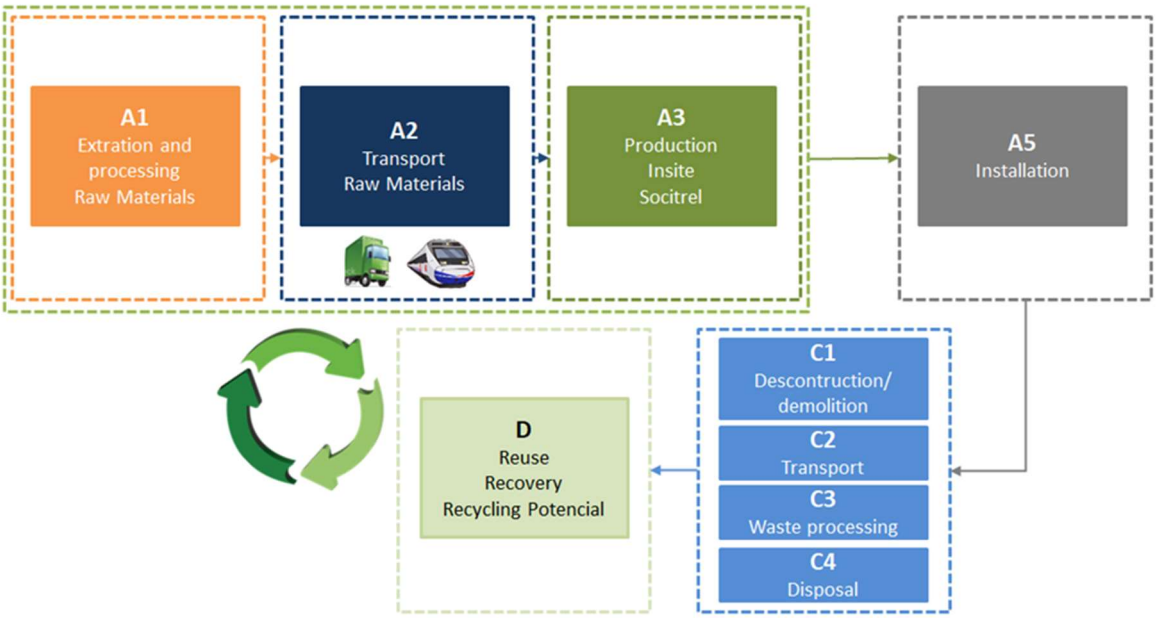
Database(s) and LCA software used:

Database: Ecoinvent v3.9.1
LCA Software: SimaPro 9.5

Description of system boundaries:

The system boundaries are Cradle to gate (A1-A3) with options, module A5, modules C1–C4 and module D. The stage (B) is not assessed because there is no significant energy or resource use and so it was excluded.
The results of module A1-A3 shouldn't be used without considering the results of module C.

System diagram:

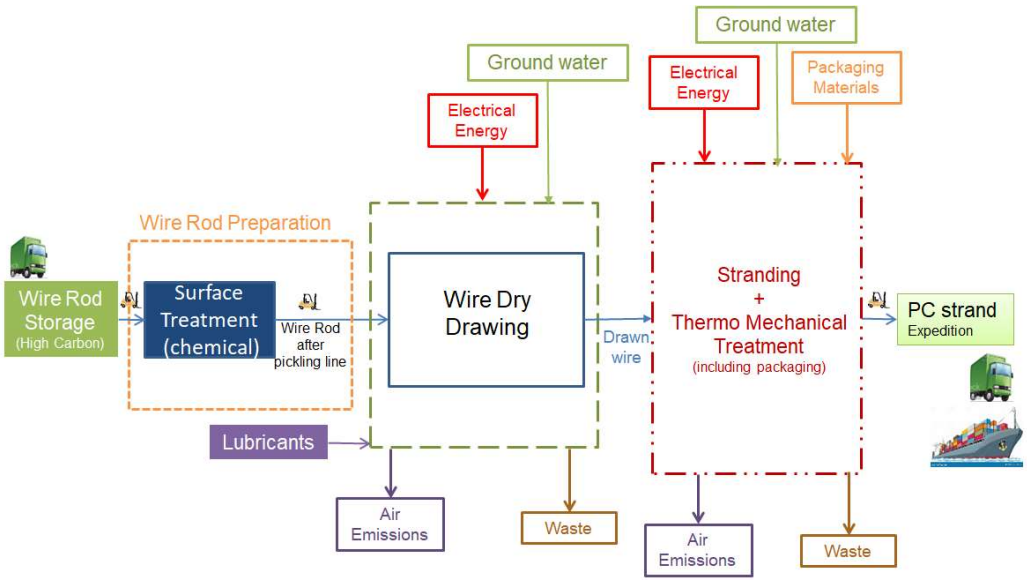


The life cycle analysis is based on the EN 15804:2012+ A2:2019 standard, where the following cutting criteria are applied:

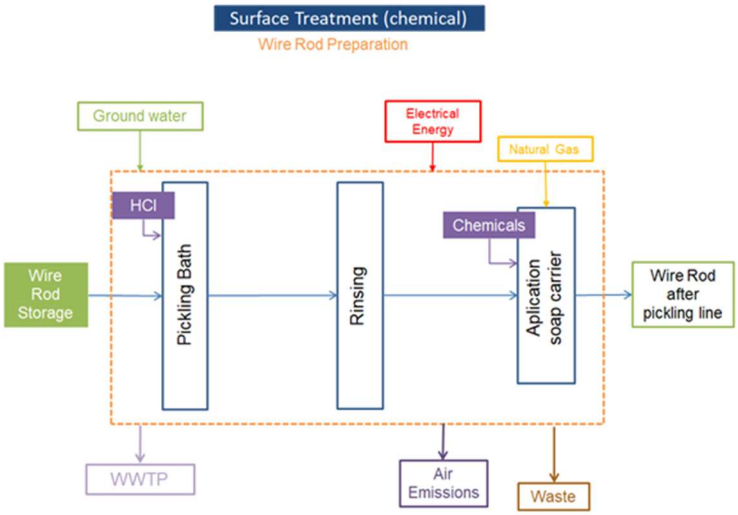
A1-A3 – Product stage

The production stage consists of the extraction of raw materials, transportation of the raw materials, processing the raw materials into materials and the production of the product. The required energy for production, external treatments, ancillary materials, packing material and production emissions are included (A1).
Then the raw materials and auxiliary material come from truck directly or by train, then by truck (A2).
The production process of the stage A3, can be described as the following:
The first step carried out at the factory is the wire rod preparation that includes chemical descalling (pickling), rinsing and application of a soap carrier.
The pickling of the wire rod is done in a batch process in each coil of wire rod is immersed in a hydrochloric acid bath in cascade, to dissolve the oxide layer in the surface. After pickling, the wire rod coils are rinsed in water, usually done in cascade rinsing also.To enhance the adhesion of the lubricant to the wire, is applied a soap carrier in the wire rod preparation, also in batch process, dipping the wire in the solution and then washing and neutralizing.
The main process in the manufacturing is the dry drawing of the wire rod, in a multipass machine. The diameter of the wire is reduced by passing it through a series of carbide dies with diminishing diameters until obtain the desired diameter of drawn wire, the cross-section could reach a medium reduction around 85%. Before entering the carbide dies, the wire is passed through a dry lubricant that reduce the friction betwen the wire and the carbide die.
The next step is the strandind process that is fed by the drawn wires. The process for 2 and 3 wires strand consists of creating a braid of 2 or 3 wires, with the same diameter. The process for 7 wires strand consists in a braid of 6 wires with the same diameter (peripherical drawn wires) around a 7th wire (central drawn wire) with a slightly larger diameter from the others (rational). After the stranding process, the strand is exposed to a thermo mechanical treatment - heated in an induction oven at approximately 400ºC and then wound on reels. The coils are then labelled, attached with steel straps and placed over wooden skids, if request, could also be wrapped with plastic-film according to the client requirement.

The production process is represented in the following diagram (for A3 stage):



The diagram for wire rod preparation:



A4 – A5 – Construction process stage

A4 - This stage is not evaluated.

A5 - Installation into the building - This stage includes the prestressed steel strand application on the construction site. In this scenario there is no energy consumption during installation, only manpower is necessary. The module A5 considers the packaging waste processing (recycling, incineration, disposal).

B- The use stage, which includes:

Stages B1-B7 not evaluated in this EPD as there are not relevant.

C1-C4 – End of life stage

The end-of-life consists of the following modules:
C1 – Deconstruction/Demolition;
C2 – Transport of waste to its Processing and end-of-life location;
C3 – Processing of waste for reuse, recovery and/or recycling (90%);
C4 – Elimination (10%).

The end-of-life is the last phase of a material's life cycle but can become the first if after demolition, recycling and reuse of waste is carried out, that is, the valorisation of the material considered to be at the end of its life.
It was assumed that 90 % of the steel is thus assumed to be recycling, and 10 % is assumed to be transported to landfill to due to low demand, according to prEN 17662:2021.

D - Recycling / reuse / recovery potential

The impacts and benefits of this stage have been included within the system boundary and, therefore, are assessed.
It was considered that 90% of the waste is recycled at end-of-life and an efficiency of 95% was taken into account in the recycling process.

More information:
LCA practitioner: Marisa Almeida / Pedro Frade – Centro Tecnológico da Cerâmica e do Vidro - Portugal
marisa@ctcv.pt

Assumptions:
For the processes to which the producers don't have any influence or specific information, such as the extraction of raw materials, generic data were used from the database Ecoinvent version 3.9.1.
The dataset used to model the production of electricity was adapted to the national reality. The electric mix was updated for the year of 2023 through the information supplied by the Nacional Energetic Network (REN), the Regulating Entity of Energetic Services (ERSE) and the Directorate-General for Energy and Geology (DGEG), to obtain the most recent results regarding the environmental impacts generated by the electric network in Portugal.
The environmental impacts presented in this EDP are related to the SOCITREL manufactured of 2023.

Criteria for the exclusion:
According to the point 6.3.5. of the NP EN 15804, the criteria for the exclusion for unitary processes is of 1% of the total energy consumed and of 1% of the total mass of the entries, with special attention for what doesn't exceed a total of 5% of the energy and mass flows excluded in the product stage.
The following process weren't considered in this study, since they can be covered by the exclusion criteria or by the norm scope:

- Environmental loads associated to the construction of industrial infrastructures and manufacturing of machinery and equipments;
- Environmental loads related to the infrastructures (production and maintenance of vehicles and roads) of transport of pre-products;
- Long-term emissions.

Allocations:
The manufacturing plant where these prestressed steel strand products are produced also produces other products, namely prestressed steel wire, railroad sleepers, annealed wire, zinc-plated wire and PVC coated wire for fences, cables, and more. Taking it into account, an mass allocation methodology was used to define which input and output flows associated only to the production of the prestressed steel strand being studied.

Data quality:
Data has been evaluated through a data quality matrix based on the EN 15804+A2+ AC (annex E) for the data quality management (Geographical, Technical and Time representativeness).
As a result of the data quality matrix, it is quantified that the gathered data achieves a good (4) level of quality (out of 5) in a range of very poor (1), poor (2), fair (3), good (4) and very good (5).

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	ND	X	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	EU	EU	EU		EU								EU	EU	EU	EU	EU
Specific data used	> 90% GWP			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	Each product declared separately (< 10%)			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	Not applicable, since the manufactured product represents only a single industrial unit.			-	-	-	-	-	-	-	-	-	-	-	-	-	-

Content information

Product components	Weight, %	Post-consumer material, weight-%	Biogenic material, weight-% and kg C/kg
Steel Wire	100	40.24 (average from several suppliers)	0
TOTAL	100	40.24 (average from several suppliers)	0
Packaging materials	**Weight, ton**	**Weight-% (versus the product)**	**Weight biogenic carbon, kg C/ton**
Wood	1.20E+00	0.12	4.19E-01
Plastic	6.53E-01	0.07	0
Raffia	2.20E-01	0.02	0
Steel packaging	3.30E+00	0.33	0
TOTAL	5.37E+00	0.54	4.19E-01

Dangerous substances from the candidate list of SVHC for Authorisation	EC No.	CAS No.	Weight-% per functional or declared unit
NA	-	-	-

The product does not contain any candidate substance from the REACH candidate list of substances of very high concern with more than 0,1% by mass.

Results of the environmental performance indicators

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory impact category indicators according to EN 15804 (version EF 3.0)

Results per functional or declared unit (per 1 ton of prestressed steel strand)								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-fossil	kg CO_2 eq.	1.05E+03	1.39E+00	4.76E+01	2.70E+01	2.12E+01	5.73E-01	-8.36E+02
GWP-biogenic	kg CO_2 eq.	-3.33E+01	5.22E+01	1.27E-02	2.16E-02	4.79E-01	9.14E-03	-3.01E+00
GWP-luluc	kg CO_2 eq.	2.51E+00	1.11E-05	1.91E-03	2.05E-04	1.63E-02	2.22E-04	-1.80E-01
GWP-total	kg CO_2 eq.	1.02E+03	5.36E+01	4.77E+01	2.70E+01	2.17E+01	5.83E-01	-8.39E+02
ODP	kg CFC 11 eq.	3.06E-05	4.64E-08	7.34E-07	6.27E-06	3.69E-07	7.50E-09	-1.75E-05
AP	mol H^+ eq.	3.58E+00	1.39E-03	4.46E-01	5.32E-02	1.38E-01	4.75E-03	-2.76E+00
EP-freshwater	kg P eq.	3.69E-02	5.76E-07	3.98E-05	1.47E-05	3.25E-04	6.16E-06	-3.81E-02
EP-marine	kg N eq.	7.76E-01	6.11E-04	2.09E-01	8.87E-03	5.89E-02	2.05E-03	-5.99E-01
EP-terrestrial	mol N eq.	8.63E+00	6.23E-03	2.28E+00	9.88E-02	6.38E-01	2.23E-02	-6.89E+00
POCP	kg NMVOC eq.	2.98E+00	1.65E-03	5.47E-01	3.49E-02	1.54E-01	5.45E-03	-2.27E+00
ADP - minerals&metals*	kg Sb eq.	6.46E-04	1.81E-08	1.96E-06	1.15E-06	9.22E-07	2.13E-08	-1.54E-04
ADP-fossil*	MJ	1.22E+04	2.92E+00	6.13E+02	3.83E+02	3.04E+02	7.18E+00	-8.09E+03
WDP*	m^3	3.70E+02	5.15E-02	7.89E-01	-8.11E-02	1.02E+00	2.47E-02	-1.66E+02
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption							

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

Additional mandatory and voluntary impact category indicators

Results per functional or declared unit (per 1 ton of prestressed steel strand)								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	1.05E+03	1.39E+00	4.76E+01	2.70E+01	2.12E+01	5.74E-01	-8.36E+02
Potential incidence of disease due to PM emissions PM	Disease incidence	9.35E-05	4.67E-08	1.24E-05	1.56E-06	2.98E-06	1.22E-07	-5.15E-05
Potential Human exposure efficiency relative to U235 IRP	kBq U 235 eq.	2.84E+01	1.26E-02	7.21E-02	1.69E+00	6.93E-01	6.40E-03	-8.03E+00
Potential Comparative Toxic Unit for ecosystems ETP-fw	CTUe	2.34E+03	4.81E+00	2.44E+02	1.53E+02	1.00E+02	2.45E+00	-1.32E+03
Potential Comparative Toxic Unit for humans. cancer effects HTP-c	CTUh	4.25E-06	6.10E-10	1.80E-09	2.02E-09	1.88E-09	2.97E-11	-3.83E-06
Potential Comparative Toxic Unit for humans. not cancer effects HTP-nc	CTUh	1.52E-05	7.79E-09	2.00E-07	2.39E-07	7.75E-08	3.46E-09	-1.38E-05
Potential soil quality index SQP	--	5.65E+03	1.23E-01	1.44E+00	1.22E+00	1.02E+01	1.75E+01	-9.19E+02

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Resource use indicators (option A).

Results per functional or declared unit (per 1 ton of prestressed steel strand)									
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**	**A-C**
PERE	MJ	2.53E+03	3.75E+01	1.19E+00	5.63E-01	1.45E+01	1.76E-01	-2.18E+02	2.58E+03
PERM	MJ	1.87E+01	-1.87E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	2.55E+03	1.87E+01	1.19E+00	5.63E-01	1.45E+01	1.76E-01	-2.18E+02	2.58E+03
PENRE	MJ	1.31E+04	7.49E+01	6.51E+02	4.06E+02	3.22E+02	7.64E+00	-8.53E+03	1.46E+04
PENRM	MJ	7.18E+01	-7.18E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	1.32E+04	3.11E+00	6.51E+02	4.06E+02	3.22E+02	7.64E+00	-8.53E+03	1.46E+04
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	3.70E+02	5.15E-02	7.89E-01	-8.11E-02	1.02E+00	2.47E-02	-1.66E+02	3.72E+02
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water								

Waste indicators

Results per functional or declared unit (per 1 ton of prestressed steel strand)								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	9.11E-02	7.70E-06	4.09E-03	1.01E-03	1.62E-03	4.03E-05	-8.36E-02
Non-hazardous waste disposed	kg	1.66E+02	2.60E-01	4.52E-02	1.60E-02	1.82E-01	1.00E+02	-1.06E+02
Radioactive waste disposed	kg	1.60E-02	2.02E-05	2.98E-05	2.77E-03	5.45E-04	4.03E-06	-6.05E-03

Output flow indicators

Results per functional or declared unit (per 1 ton of prestressed steel strand)								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Material for recycling	kg	6.82E+00	0.00E+00	0.00E+00	0.00E+00	9.00E+02	0.00E+00	0.00E+00
Materials for energy recovery	kg	7.47E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy. electricity	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy. thermal	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Other environmental performance indicators

Not applicable.

Additional environmental information

Not applicable.

Additional social and economic information

Not applicable.

Information related to Sector EPD

Not applicable. This is an individual EPD.

Differences versus previous versions

There is no previous EPD.

References

General Programme Instructions of the International EPD® System. Version 4.0.

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ISO 14025:2010 Environmental labels and declarations - Type III environmental declarations - Principles and procedures.

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